SET 5: THE FENSAL FLATS ATMOSPHERIC XENON & NITROGEN SIPHONING ARRAY

DOCUMENT 5A: SECURED CREDIT AGREEMENT

SECURED TERM CREDIT AGREEMENT

Dated as of the Agreement Effective Date (as defined herein)

among

FENSAL ATMOSPHERE HARVESTING CORP,
as Borrower

and

JOVIAN-SATURNINE INDUSTRIAL CONSORTIUM (JSIC),
as Lead Arranger, Administrative Agent, and Collateral Agent

and

THE LENDERS PARTY HERETO FROM TIME TO TIME

This SECURED TERM CREDIT AGREEMENT (this "Agreement") is entered into on the Agreement Effective Date by and among the Borrower, the Lenders, and the Administrative Agent. In consideration of the mutual covenants and agreements herein contained, the parties hereto agree as follows:

ARTICLE I: DEFINITIONS AND INTERPRETATION

Section 1.01. Defined Terms. As used in this Agreement, the following terms have the meanings specified below:

- **"Agreement Effective Date"** means the date on which this Agreement has been fully executed and delivered by all parties hereto, as verified by the date inscribed next to the signatures of the authorized officers in the execution block below (such date, the **"Signature Date"**).

- **"Atmospheric Ballast Escrow"** means the cash escrow reserve fund mandated to be maintained under Section 4.01(3) for secondary nitrogen ballast regulation and localized pressure containment.

- **"Siphoning Array"** means the physical structural harvesting towers, high-altitude compression intakes, vacuum refrigeration ducts, and noble gas fractional condensation units located at Sector 15, Fensal Flats coordinates (latitude 5.2° N, longitude 30.1° W) on Titan, Saturn.

- **"Sol System Interstellar Accounting Standards"** or **"SSIAS"** means the unified framework of financial accounting principles approved by the Interstellar Financial Board.

- **"TIDOR"** means the Titan Inter-Dome Offered Rate, which is the floating interbank benchmark rate calculated for Titan-denominated credit lines.

- **"Xenon Siphoning Efficiency Index"** or **"XSEI"** means the noble gas extraction and molecular separation efficiency rating of the primary fractionation column, as certified on a 100-point basis by the Titan Technical Safety Directorate (TTSD).

- **"Xenon Yield Target"** means the minimum operational noble gas recovery metric of 4,500 kilograms of pure liquid Xenon (Xe) per Titan sidereal day, as calculated under the provisions of Section 4.01(2) herein.

Section 1.02. Rules of Interpretation. Capitalized terms defined in the singular shall have the corresponding meanings in the plural, and vice versa. Any references to specific sections, tables, or documents relate directly to this Agreement unless otherwise specified.

ARTICLE II: THE SECURED CREDIT LINES

Section 2.01. The Facility. Subject to the terms and conditions of this Agreement, each Lender agrees to make a single advance term loan (each a "Facility Loan") to the Borrower on the **Closing Date** (which is identical to the **Signature Date**) in an aggregate principal amount of **5,100,000,000 Sol Credits (SC)**.

Section 2.02. Interest Rate Adjustments.

1. The Facility Loans shall accrue interest daily at a floating rate per annum equal to **TIDOR plus 6.25%**.
2. Should the **Xenon Siphoning Efficiency Index (XSEI)** fall below 91.0% for any consecutive 5-day cycle, the floating margin shall automatically increase by **2.75% per annum** (the "Xenon Venting Penalty") until the XSEI is restored to 94.0% or above, as verified by the TTSD.

Section 2.03. Repayment and Amortization. Repayment of the principal shall be executed in equal quarterly installments on the final operational day of each Martian fiscal quarter, starting on the first such date following the **Agreement Effective Date**, in accordance with the amortization schedules appended hereto.

ARTICLE III: REPRESENTATIONS AND WARRANTIES

The Borrower represents and warrants to the Lenders that:

Section 3.01. Atmospheric Harvesting Rights. The Borrower holds exclusive licenses to harvest, compress, and process atmospheric compounds within Sector 15-Fensal, which contains optimal concentrations of industrial-grade nitrogen, argon, and high-value noble gas trace compounds under the authority of the Titan Regulatory Council (TRC).

Section 3.02. Wind-Shear Tolerance. The structural towers of the Siphoning Array are engineered using dynamic tension carbon-fiber guy-wires and active gyroscope dampers capable of neutralizing heavy tropospheric wind-shear forces up to 150 meters per second.

ARTICLE IV: COVENANTS

Section 4.01. Affirmative Covenants. So long as any part of the Facility Loan remains outstanding, the Borrower shall:

1. **Siphon Array Maintenance:** Undertake continuous acoustic-resonance structural audits of the primary intake pylons to ensure the XSEI does not fall below the base standard of 92.0%.
2. **Maintain Xenon Yield:** Operate the condensation columns to sustain the **Xenon Yield Target** of 4,500 kilograms per Titan day, ensuring supply covenants with inter-planetary ion-propulsion transport vessels are satisfied.
3. **Atmospheric Escrow:** Maintain the cash-funded **Atmospheric Ballast Escrow** with the Administrative Agent, holding a balance of not less than SC 350,000,000 at all times.

Section 4.02. Negative Covenants. The Borrower shall not, without the explicit written approval of the Administrative Agent:

1. **Tower Modifications:** Alter, depressurize, or adjust the intake flow parameters of any active fractionation columns within the designated coordinates.

2. **Restricted Payments:** Declare or make any dividend distributions to corporate parent entities if the XSEI is below 93.0% or if a Default Event has occurred and is continuing.

ARTICLE V: DEFAULT EVENTS AND REMEDIES

Section 5.01. Default Events. The occurrence of any of the following events shall constitute an Event of Default:

1. **Wind-Shear Structural Collapse:** Any catastrophic structural failure or partial collapse of the siphoning towers due to extreme meteorological atmospheric pressure shifts or localized storm cycles.

2. **Covenant Non-Compliance:** Failure to maintain the required balance in the **Atmospheric Ballast Escrow** under Section 4.01(3), or failure to meet the **Xenon Yield Target** for more than two consecutive quarters.

3. **Uncontained Nitrogen Blowout:** Any pressurized piping failure leading to an explosive decompression event discharging more than 150,000 standard cubic meters of compressed nitrogen into the active dome habitation zones.

FENSAL ATMOSPHERE HARVESTING CORP

By: _____
Name: Alistair Crane-Vance
Title: Managing Director, Gas Extraction Operations

Date of Execution: **June 26, 2146**

JOVIAN-SATURNINE INDUSTRIAL CONSORTIUM

By: _____
Name: Diana Ross-Vargas
Title: Senior Partner, Inter-Planetary Infrastructure Fund

Date of Execution (the **Signature Date**): **June 26, 2146**